

SMC PLIC — Test Plan

SMC PLIC — Detailed Test Plan

Document: 03_PLIC_Test_Plan.md **Module under test (MUT):** smc::plic (peripherals/plic/) **Reference test bench:** peripherals/plic/test/plic_tb.cpp **Status:** Active — all 20 unit tests pass on SystemC 3.0.2 / Apple Clang 17 **Companion docs:** - 01_PLIC_Specification.md — externally-observable behaviour - 02_PLIC_LowLevel_Design.md — implementation contract

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1. Objectives

The PLIC test plan must demonstrate that the SystemC model:

1. **Conforms** to the RISC-V PLIC v1.0 specification and the OCAH plic.rdl register layout.
 2. **Implements correctly** every flow described in 01_PLIC_Specification.md §7 (source latching, arbitration, claim/ complete, threshold gating, cross-context isolation, reset).
 3. **Honours** the TLM-2.0 contract documented in 01_PLIC_Specification.md §9 (alignment, width, error responses, debug back door).
 4. **Is robust** against negative stimulus (misaligned, oversize, out-of-window, spurious EOI, bad commands).
 5. **Is integration-ready** — the unmodified firmware driver (fw/smc/common/drivers/riscv_plic0.c) runs against it through the full fabric path.
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2. Scope (in / out)

2.1 In scope

- Self-checking SystemC unit tests for every register and every observable behavioural rule.
- Test-bench-driven stimulus through `b_transport` (the production access path).
- Back-door inspection through `dbg_*` and `transport_dbg`.
- Negative path testing (misaligned, oversize, out-of-window, bad commands).
- Reset behaviour, including from a non-trivial state.

2.2 Out of scope (covered by other plans)

Aspect	Owner / document
Cycle-accurate timing of MEIP/SEIP delivery	RTL verification (Chipyard)
Access-control / <code>axi_filter</code> enforcement	<code>axi_filter</code> test plan
Multi-master arbitration on the SMC fabric	<code>smc_fabric</code> test plan
<code>riscv_plic0.c</code> driver-level scenarios	SMC firmware test plan
Whisper ↔ Rocket commit-log diff	Co-simulation test plan (future)
Power / DFT / MBIST	RTL DV plan

3. Verification strategy

3.1 Tier model

Tier	Goal	Where
Unit (this plan)	Per-register R/W, per-flow correctness	<code>plic_tb.cpp</code>
Integration	Fabric → filter → PLIC → CPU cluster	<code>smc_top</code> integration test (future)
System	Firmware running real PLIC driver under Whisper	<code>smc_top</code> boot test (future)
Co-simulation	Whisper vs Rocket commit-log diff with PLIC events	future

3.2 Methods

Method	Purpose
Directed tests	Trigger a specific spec requirement and check the observable response.
Back-door assertions (<code>dbg_*</code>)	Inspect internal state with no perturbation.
Stress against in-window holes	Confirm reserved bits decode to RAZ/WI rather than corrupting state.
Negative stimulus	Confirm the bus-error contract holds for misaligned / oversize / OOR.
Reset replay	Drive non-trivial state, then assert reset, then re-check from scratch.
Two-phase settling	After every state mutation, run two delta cycles to let <code>recompute_event_</code> propagate before checking outputs.

3.3 Determinism guarantees

The test bench is fully deterministic:

- No random stimulus.
- No wall-clock dependence; settles via `wait(SC_ZERO_TIME)` instead of real time.
- Single-threaded SystemC scheduler, single TLM initiator.

A failing test is a hard error — no retries, no flaky regression flags.

4. Environment

Component	Required version
Accellera SystemC	≥ 2.3.4 (3.0.2 verified)
OSCI CCI	cci-1.0.0 (Accellera reference implementation)
C++ compiler	C++17 — Apple Clang 17 / Clang 10+ / GCC 9+
CMake	≥ 3.16
OS (host)	macOS / Linux
Optional	ctest for CI integration (already wired)

Reproducible commands:

```
SYSTEMC_HOME=/opt/homebrew/opt/systemc \  
  cmake -S peripherals/plic -B peripherals/plic/build -DCMAKE_BUILD_TYPE=Release  
cmake --build peripherals/plic/build -j  
ctest --test-dir peripherals/plic/build --output-on-failure
```

Or, equivalently:

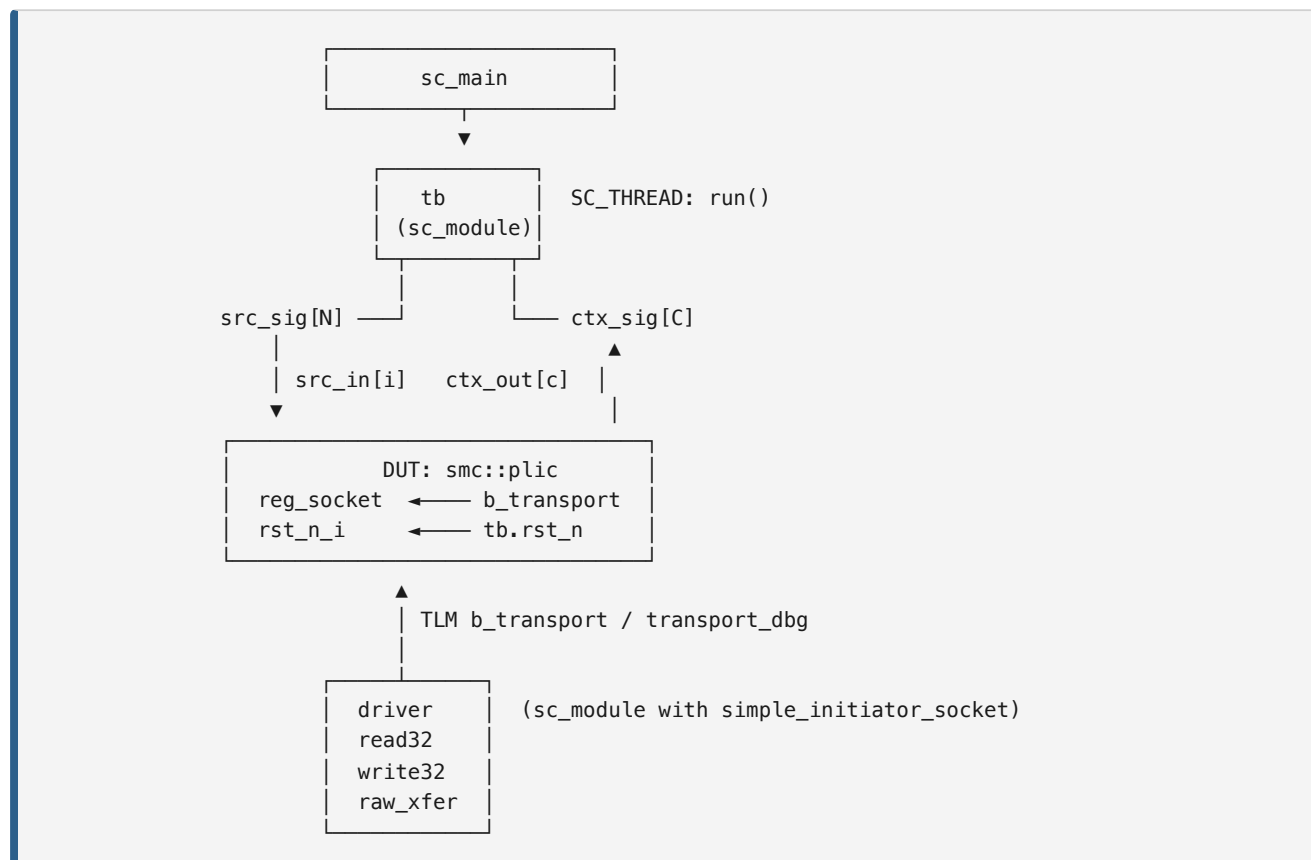
```
peripherals/plic/run_tests.sh           # configure (if needed) + build + run  
peripherals/plic/run_tests.sh --clean  # wipe build/ first  
peripherals/plic/run_tests.sh --ctest  # run via ctest with PASS/FAIL regex
```

The CMake CTest entry asserts on the regex:

```
PASS_REGULAR_EXPRESSION "ALL TESTS PASSED"  
FAIL_REGULAR_EXPRESSION "FAIL"
```

so any test bench run that does not print `ALL TESTS PASSED` is a hard CI failure.

5. Test bench architecture



`tb` instantiates:

- `dut` — the PLIC under test, with default config (336 sources, 8 contexts).
- `drv` — a TLM initiator that exposes `read32` / `write32` / `raw_xfer`.
- `rst_n` and `sc_vector<sc_signal<bool>>` arrays for sources and outputs.

The `run()` thread executes the test sequence; on completion it calls `sc_stop()` and returns the failure count to the OS.

6. Stimulus & checking primitives

Primitive	Purpose
<code>EXPECT_EQ(a, b)</code>	Hard equality check; logs <code>FAIL</code> on mismatch and increments <code>g_failures</code> .
<code>EXPECT_TRUE(c)</code>	Boolean-truth check; same failure semantics as <code>EXPECT_EQ</code> .
<code>pulse_reset()</code>	Drives <code>rst_n</code> low for 20 ns, then high for 20 ns, then yields enough delta cycles for state to settle.
<code>raise(src, level)</code>	Drives <code>src_sig[src - 1]</code> to the requested level and settles.
<code>settle()</code>	Two <code>wait(SC_ZERO_TIME)</code> calls — enough for <code>src_method/b_transport</code> → <code>recompute_event_</code> → <code>output_method</code> to propagate to <code>ctx_out</code> .
<code>drv.read32(addr)</code>	TLM read; auto-fails on non- <code>TLM_OK_RESPONSE</code> .
<code>drv.write32(addr, value)</code>	TLM write; auto-fails on non- <code>TLM_OK_RESPONSE</code> .
<code>drv.raw_xfer(cmd, addr, len, data)</code>	Returns the raw <code>tlm_response_status</code> for negative-path tests.

Address arithmetic helpers:

```

prio_addr(src)          = PRIORITY_BASE + 4 * src
pending_addr(word)      = PENDING_BASE  + 4 * word
enable_addr(ctx, word)  = ENABLE_BASE   + ENABLE_STRIDE * ctx + 4 * word
thr_addr(ctx)           = CONTEXT_BASE   + CONTEXT_STRIDE * ctx + CONTEXT_THR_OFF
cc_addr(ctx)            = CONTEXT_BASE   + CONTEXT_STRIDE * ctx + CONTEXT_CC_OFF

```

7. Feature → test traceability matrix

#	Specification requirement (01_..._Specification.md)	Verified by
F1	Reset clears priority / enable / threshold / pending / claim_in_flight	TC-1, TC-14
F2	PRIORITY[*] is 3-bit (writes truncated)	TC-2
F3	Source 0 is reserved (RAZ/WI on PRIORITY)	TC-3
F4	Pending bit set on rising edge of source line	TC-4
F5	PENDING register reflects per-source pending state, packed 32 per word	TC-4
F6	Disabled (enable=0) source does not drive ctx_out	TC-5
F7	Per-context enable independence	TC-6, TC-13
F8	Threshold gates priority strictly (priority > threshold)	TC-7
F9	Best-pending arbitration — highest priority wins	TC-8
F10	Tie-break — lowest source ID wins	TC-8
F11	Claim returns top-pending, clears pending, sets claim_in_flight	TC-9
F12	Complete with line still high re-arms pending immediately	TC-9
F13	Complete after line de-asserted does not re-arm pending	TC-10
F14	Misaligned access → TLM_BURST_ERROR_RESPONSE	TC-11
F15	Oversize access → TLM_BURST_ERROR_RESPONSE	TC-11
F16	Out-of-window access → TLM_ADDRESS_ERROR_RESPONSE	TC-11
F17	transport_dbg claim-read does not claim	TC-12
F18	Per-context threshold independence (raising one ctx threshold does not affect others)	TC-13
F19	Reset returns to clean state from non-trivial state	TC-14
F20	Source 0 is never returned as a claim result	Implicit in TC-8/TC-9 (best_pending iterates from 1)
F21	DMI hint always cleared	Constructor/ b_transport invariant — see 02_..._LowLevel_Design.md §4.2
F22	complete() of out-of-range source ID is silently dropped	Implicit in spec; covered by complete() precondition
F23	dbg_priority / dbg_enable / dbg_threshold / dump_state return correct values and safe defaults for OOB inputs	TC-15
F24	Write to PENDING region is silently discarded (SW=r)	TC-16
F25	Context-block reserved sub-offset reads as 0 (RAZ) and writes are ignored (WI)	TC-17
F26	Unknown TLM command → TLM_COMMAND_ERROR_RESPONSE	TC-18
F27	transport_dbg general read (non-CC) and write paths function correctly	TC-19
F28	Enable last-word bits beyond num_sources are forced to 0 on write	TC-20

Spec features F1–F22 were exercised before the TC-15..TC-20 additions; F23–F28 are the newly covered requirements.

8. Detailed test cases (TC-1 ... TC-20)

Each test case lists: *Spec ref* → *Setup* → *Stimulus* → *Expected* → *Notes*.

TC-1. Reset clears all state

- **Spec:** F1
- **Setup:** Default `plic_cfg` (336 sources, 8 contexts).
- **Stimulus:** Drive `rst_n` low for 20 ns, then high; allow settle.
- **Expected:**
 - `read32(prio_addr(s))` returns 0 for every $s \in [1..N]$.
 - `read32(thr_addr(c))` returns 0 for every $c \in [0..C-1]$.
 - `ctx_sig[c]` reads false for every c .
- **Notes:** Reads are issued through the bus to confirm the register bank, not just internal state, is cleared.

TC-2. Priority R/W with 3-bit truncation

- **Spec:** F2
- **Setup:** After reset.
- **Stimulus:**
 - `write32(prio_addr(7), 0x12345678)` — verify `[2:0]` survives, upper bits drop.
 - `write32(prio_addr(7), 5)` — verify exact value reads back.
- **Expected:** First read returns `0x12345678 & 0x7 == 0x0`; second read returns 5.

TC-3. Source 0 reserved

- **Spec:** F3
- **Setup:** After reset.
- **Stimulus:** `write32(prio_addr(0), 7)`, then `read32(prio_addr(0))`.
- **Expected:** Read returns 0 (RAZ; write was silently ignored).

TC-4. Pending latched on rising edge

- **Spec:** F4, F5
- **Setup:** Source 7 not yet pending.
- **Stimulus:** `raise(7, true)` — drives `src_in[6]` high.
- **Expected:**
 - `dbg_pending(7)` is true.
 - Reading the pending word `pending_addr(7/32) == 0x1000`, bit 7 & 31 == 7 is set.

TC-5. Disabled source does not drive `ctx_out`

- **Spec:** F6
- **Setup:** Continuation of TC-4 — pending without enable.
- **Stimulus:** settle.

- **Expected:** `ctx_sig[c]` is false for every `c`.

TC-6. Per-context enable independence

- **Spec:** F7
- **Setup:** Source 7 pending.
- **Stimulus:** `write32(enable_addr(0, 0), 1u << 7)` — enable in ctx 0 only.
- **Expected:** `ctx_sig[0]` rises; `ctx_sig[1..7]` stay low.

TC-7. Threshold gating

- **Spec:** F8
- **Setup:** Source 7 priority = 5, enabled in ctx 0.
- **Stimulus:**
 - `write32(thr_addr(0), 6) → 5 ≤ 6 → masked.`
 - `write32(thr_addr(0), 4) → 5 > 4 → visible.`
- **Expected:** `ctx_sig[0]` falls then rises accordingly.

TC-8. Best-pending arbitration (priority + tie-break)

- **Spec:** F9, F10
- **Setup:** Sources 7 / 11 / 13 enabled in ctx 0; thresholds = 4.
- **Stimulus:**
 - `priority[7] = 5, priority[11] = 6, priority[13] = 5.`
 - Raise sources 11 and 13.
 - Read `dbg_claim_top(0) → expect 11 (higher priority).`
 - Set `priority[11] = 0 → expect dbg_claim_top(0) returns 7 (lowest ID at priority 5 — 7 < 13).`

TC-9. Claim/complete with line still high

- **Spec:** F11, F12
- **Setup:** Source 7 priority 5, enabled in ctx 0, line high; source 13 silenced via priority 0.
- **Stimulus:**
 - `read32(cc_addr(0)) → claim → returns 7.`
 - `write32(cc_addr(0), 7) → complete with line still high.`
- **Expected:**
 - Claim returns 7, `dbg_pending(7)` becomes false, `ctx_sig[0]` falls.
 - After complete, `dbg_pending(7)` becomes true again, `ctx_sig[0]` rises.

TC-10. Complete after line de-asserted — no re-pend

- **Spec:** F13
- **Setup:** Continuation of TC-9.
- **Stimulus:**
 - `read32(cc_addr(0)) → claim again → returns 7.`

- `raise(7, false)` → drop the line.
- `write32(cc_addr(0), 7)` → complete.
- **Expected:** `dbg_pending(7)` stays false; `ctx_sig[0]` stays low.

TC-11. TLM error responses

- **Spec:** F14, F15, F16
- **Stimulus** (via `raw_xfer`):
 - Read at address `0x2` (misaligned, length 4) → `TLM_BURST_ERROR_RESPONSE`.
 - Read at address `0x0` (length 8) → `TLM_BURST_ERROR_RESPONSE`.
 - Read at address `WINDOW_SIZE + 4` → `TLM_ADDRESS_ERROR_RESPONSE`.

TC-12. `transport_dbg` has no side effects

- **Spec:** F17
- **Setup:** Source 7 pending, enabled, threshold 0 — claim would return 7.
- **Stimulus:** Issue a `transport_dbg` read of `cc_addr(0)`.
- **Expected:**
 - The debug call returns 4 bytes with value 7.
 - `dbg_pending(7)` remains true (no claim happened).

TC-13. Cross-context isolation

- **Spec:** F7, F18
- **Setup:** Source 7 enabled and above threshold in ctx 0.
- **Stimulus:**
 - `write32(enable_addr(4, 0), 1u << 7)` and `write32(thr_addr(4), 0)` → ctx 4 also fires.
 - `write32(thr_addr(4), 7)` → only ctx 4 mutes.
- **Expected:** `ctx_sig[4]` follows the threshold change; `ctx_sig[0]` stays high throughout.

TC-14. Reset returns to clean state from non-trivial state

- **Spec:** F1, F19
- **Setup:** After all previous tests have left the model in a multi-source, multi-context configured state.
- **Stimulus:** `pulse_reset()`.
- **Expected:**
 - All `ctx_sig[c]` low.
 - `read32(prio_addr(7)) == 0`, `read32(enable_addr(0, 0)) == 0`.

TC-15. Debug back-door API

- **Spec:** F23
- **Setup:** Post TC-14 reset (clean state). Write `priority[3]=5`, `enable[0][word0]=1<<3`, `threshold[0]=2` via `b_transport`.
- **Stimulus:** Call all four back-door helpers with valid and out-of-range / reserved inputs.
- **Expected:**

- `dbg_priority(3)` returns 5; `dbg_priority(0)` returns 0 (src 0 reserved); `dbg_priority(num_sources+1)` returns 0 (OOB).
- `dbg_enable(0, 3)` returns true; `dbg_enable(num_contexts, 3)` returns false (ctx OOB); `dbg_enable(0, 0)` returns false (src 0); `dbg_enable(0, num_sources+1)` returns false (src OOB).
- `dbg_threshold(0)` returns 2; `dbg_threshold(num_contexts)` returns 0 (ctx OOB).
- `dump_state(oss)` produces non-empty output without crashing.

TC-16. PENDING write silently discarded

- **Spec:** F24
- **Setup:** After TC-14 reset (no pending bits set).
- **Stimulus:** `write32(pending_addr(0), 0xFFFFFFFF)`.
- **Expected:** `read32(pending_addr(0))` returns 0 — write had no effect.

TC-17. Context-block reserved sub-offset (RAZ/WI)

- **Spec:** F25
- **Setup:** Any state.
- **Stimulus:**
 - `read32(CONTEXT_BASE + 0x8)` — sub-offset 8, neither THR nor CC.
 - `write32(CONTEXT_BASE + 0x8, 0xDEAD)`.
 - `read32(CONTEXT_BASE + 0x8)` again.
- **Expected:** Both reads return 0 (RAZ); write completes with `TLM_OK_RESPONSE` (WI, no error).

TC-18. `b_transport` — `TLM_COMMAND_ERROR_RESPONSE`

- **Spec:** F26
- **Stimulus:** `raw_xfer(TLM_IGNORE_COMMAND, 0x0, 4, &scratch)`.
- **Expected:** Response status is `TLM_COMMAND_ERROR_RESPONSE`.
- **Notes:** Previous TC-11 only checked BURST and ADDRESS errors.

TC-19. `transport_dbg` general read and write paths

- **Spec:** F27
- **Setup:** `priority[3] = 5` (set in TC-15).
- **Stimulus:**
 - `transport_dbg` READ of `prio_addr(3)` (non-CC address).
 - `transport_dbg` WRITE of `prio_addr(5)` with data 6.
- **Expected:**
 - Read returns 4 bytes with value 5 (`priority[3]`).
 - Write returns 4; `dbg_priority(5)` confirms the new value is 6.
- **Notes:** TC-12 only exercised the CC-peek path.

TC-20. Enable last-word bit masking

- **Spec:** F28

- **Setup:** Default 336-source config: W=11 words (0..10); last word is word 10, containing sources 320–336 (17 valid bits [0..16]).
- **Stimulus:** `write32(enable_addr(0, 10), 0xFFFFFFFF)`.
- **Expected:** `read32(enable_addr(0, 10))` returns `0x0001FFFF` (bits 17–31 forced to 0 by the write mask; `valid_bits = 337 - 10×32 = 17`).

9. Coverage plan

9.1 Functional coverage (instrumentation roadmap)

The current test bench is directed; coverage instrumentation is planned but not yet plumbed into the project's coverage harness. Hooks expected in follow-up commits:

Bin / cross	Implementation hook
Per-register R / W coverage	Increment counters at <code>reg_read</code> / <code>reg_write</code> exit
Per-source rising edges	Counter in <code>src_method</code>
Per-source <code>claim_in_flight</code> set/clear	Counters in <code>claim()</code> / <code>complete()</code>
Per-context <code>ctx_out</code> rising / falling transitions	Counters in <code>output_method</code>
Cross: $(\text{priority} \times \text{threshold}) \in \{0..7\}^2$	Sample in <code>output_method</code> after each recompute
Cross: $(\text{claim_in_flight} \times \text{line_high_at_complete})$	Sample in <code>complete()</code>
Reserved-region accesses	Counter in <code>reg_read</code> / <code>reg_write</code> reserved branch

9.2 Code coverage targets and current results

Both `--asan` and `--coverage` modes are wired into `run_tests.sh` and use separate build directories (`build_asan/`, `build_cov/`). Coverage uses LLVM source-based instrumentation (Clang) or `--coverage` / `gcov` (GCC), selected automatically by CMake based on `CMAKE_CXX_COMPILER_ID`.

Metric	Target	Current (TC-1..TC-20)
Function coverage	100 %	100 % (20/20)
Line coverage	≥ 95 %	97.7 % (293/300)
Region coverage	≥ 80 %	90.4 %

The 7 lines that remain uncovered are structurally unreachable from the test bench: - 4 `SC_REPORT_FATAL` constructor guard lines (require a fatal abort to trigger; see TC-plan future item N13/N14). - 3 `return false` / `return 0` defensive paths that are pre-empted by the `b_transport` window check before `reg_read` / `reg_write` can see an out-of-window address.

Untestable lines are candidates for `// LCOV_EXCL_LINE` annotations.

9.3 Coverage closure plan

A test is considered insufficient if any spec requirement in §7 is covered by zero tests, or any covergroup bin in §9.1 has count 0 after the regression. Closure is reached when:

- All spec rows (F1..F22) have at least one passing test.
- All coverage bins are hit at least once.
- Negative-test catalogue (§10) is fully exercised.
- Code coverage targets in §9.2 are met.

10. Negative-test catalogue

#	Negative scenario	Expected response	Covered by
N1	Misaligned read (e.g. <code>addr = 0x2</code>)	<code>TLM_BURST_ERROR_RESPONSE</code>	TC-11
N2	Misaligned write	<code>TLM_BURST_ERROR_RESPONSE</code>	TC-11 (symmetric path)
N3	Oversize access (length $\neq$ 4)	<code>TLM_BURST_ERROR_RESPONSE</code>	TC-11
N4	Undersize access (length $<$ 4)	<code>TLM_BURST_ERROR_RESPONSE</code>	TC-11 (path)
N5	Out-of-window access ($\geq$ <code>WINDOW_SIZE</code>)	<code>TLM_ADDRESS_ERROR_RESPONSE</code>	TC-11
N6	<code>TLM_IGNORE_COMMAND</code>	<code>TLM_COMMAND_ERROR_RESPONSE</code>	TC-18 ✓
N7	<code>priority[0]</code> write	RAZ/WI	TC-3
N8	<code>enable[c][0]</code> write of bit 0	Bit 0 forced to 0	Implicit in TC-6
N9	Write to reserved bits (above 3 bits) of <code>priority[s]</code>	Truncated to 3 bits	TC-2
N10	Write to <code>pending[*]</code>	Silently discarded (RO)	TC-16 ✓
N11	<code>complete(ctx, src=0)</code>	Silently dropped	Implicit in <code>complete()</code> precondition
N12	<code>complete(ctx, src > num_sources)</code>	Silently dropped	Implicit in <code>complete()</code> precondition
N13	Construction with <code>num_sources = 0</code> or $>$ 1023	<code>SC_REPORT_FATAL</code> at elaboration	Future (planned)
N14	Construction with <code>num_contexts = 0</code>	<code>SC_REPORT_FATAL</code> at elaboration	Future (planned)
N15	DMI request	<code>dmi_allowed = false</code> (no callback registered)	Implicit (callback never registered)

Items marked `Future (planned)` are listed in §13.

11. Regression workflow

11.1 Local regression

```
peripherals/plic/run_tests.sh
```

The script:

1. Auto-locates SystemC (`SYSTEMC_HOME` or Homebrew default).
2. Configures CMake (only on first run; subsequent runs are incremental).

- 3. Builds with `-j<ncpus>`.
- 4. Runs the test bench binary `peripherals/plic/build/test/plic_tb`.

11.2 CI regression

```
peripherals/plic/run_tests.sh --ctest
```

`ctest` parses the test bench output and asserts on:

- `PASS_REGULAR_EXPRESSION "ALL TESTS PASSED"` — must match.
- `FAIL_REGULAR_EXPRESSION "FAIL"` — must not match.

The command exits non-zero on either condition.

11.3 Triage protocol

On failure:

- 1. Read the first `FAIL` line from `stderr` — it carries `__FILE__:__LINE__` plus the expected/actual values.
- 2. Optionally re-run with `--clean` to rule out a stale build.
- 3. Add `dut.dump_state(std::cerr)` adjacent to the failing assertion to capture full PLIC state.
- 4. If the failure reproduces under a single test case but not in isolation, suspect cross-test state leakage — most likely an `unreset pending_` from a prior block.

12. Pass / fail criteria & exit conditions

Outcome	Criterion
PASS	The test bench prints <code>ALL TESTS PASSED</code> and exits with status <code>0</code> .
FAIL	Any <code>EXPECT_*</code> macro reports <code>FAIL</code> , or the bench prints any <code>FAIL</code> line, or it exits with non-zero status.
BUILD FAIL	CMake configuration or compilation error — counts as <code>FAIL</code> for CI.
TIMEOUT (CI only)	Bench has not completed within 60 s — counts as <code>FAIL</code> (current actual: < 1 s).

A green CI run requires every test case TC-1..TC-20 to pass on every supported (SystemC version × compiler) matrix entry.

13. Future tests & open work

Items not yet covered by `plc_tb.cpp` but planned:

Item	Priority	Notes
Negative cmd: TLM_IGNORE_COMMAND returns TLM_COMMAND_ERROR_RESPONSE	—	Done (TC-18)
Write to pending[+] is silently discarded	—	Done (TC-16)
Construction-time SC_REPORT_FATAL for bad num_sources / num_contexts (N13/N14)	Low	CCI preset path: set a bad value via broker preset before ctor; use <code>sc_report_handler::set_actions(SC_FATAL, SC_THROW)</code> to capture
CCI preset override test — verify that a broker preset for num_sources overrides the plic_cfg default	Medium	Requires a CCI broker fixture; confirms the dual-mechanism described in <code>01_PLIC_Specification.md §4</code>
CCI mutable access_delay_ns — change between transactions and verify annotated delay shifts	Medium	Read back <code>b_transport</code> delay before and after a broker write
Source 0 explicitly never returned as claim	Low	Force <code>pending[0] = true</code> via test-only hook
Per-source coverage of (claim, line low at complete) cross	Medium	Coverage plumbing
DMI request rejection	Low	Verify the absence of a <code>get_direct_mem_ptr</code> handler
Multi-PLIC instantiation	Low	Build a TB with two <code>plic</code> modules and disjoint sources
Random-stimulus fuzzer driving register accesses + edges	Medium	Bound-checked random sequence; compare against reference model
Whisper+PLIC system test	High	Boot SMC firmware against the model through the fabric

14. Test results — current baseline

Test	Status
TC-1 reset clears state	PASS
TC-2 priority R/W + 3-bit truncation	PASS
TC-3 source 0 reserved	PASS
TC-4 pending latched on rising edge	PASS
TC-5 disabled source does not drive ctx_out	PASS
TC-6 per-context enable independence	PASS
TC-7 threshold gating	PASS
TC-8 best-pending arbitration	PASS
TC-9 claim/complete with line still high	PASS
TC-10 complete after line de-asserted	PASS
TC-11 TLM error responses	PASS
TC-12 transport_dbg has no side effects	PASS
TC-13 cross-context isolation	PASS
TC-14 reset returns to clean state	PASS
TC-15 debug back-door API	PASS
TC-16 PENDING write silently discarded	PASS
TC-17 context-block reserved sub-offset (RAZ/WI)	PASS
TC-18 TLM_COMMAND_ERROR_RESPONSE	PASS
TC-19 transport_dbg general read and write	PASS
TC-20 enable last-word bit masking	PASS

Result: 20 / 20 PASS — ALL TESTS PASSED **Environment:** SystemC 3.0.2 Accellera, Apple Clang 17, macOS 24.3.0 **Wall-clock:** < 1 second on Apple Silicon. **Coverage** (LLVM instrumented, `run_tests.sh --coverage`): function 100 % · line 97.7 % · region 90.4 %

End of document.